Homework — 1.2.1 Operating Systems

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.2.1 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. An operating system is said to provide *abstraction* and to act as a *platform* for application software. Explain what each of these two ideas means. **[2]** *(AO1)*

2. A computer is running several programs at once even though it has a single CPU core. Explain how **scheduling** creates the illusion that the programs run simultaneously, and state **two** goals a good scheduler tries to achieve. **[3]** *(AO1)*

3. Compare the **First Come First Served (FCFS)** and **Shortest Job First (SJF)** scheduling algorithms. In your answer name **one** drawback of each. **[3]** *(AO1 / AO2)*

4. A computer with 8 GB of RAM is running a video editor that needs more memory than is physically available. Explain how **virtual memory** allows the program to run, and explain why running too many large programs this way can lead to **disk thrashing**. **[4]** *(AO2)*

5. State what the **BIOS** does when a computer is first switched on. Give **two** distinct tasks it performs. **[2]** *(AO1)*

Part B — Retrieval: keep it fresh (~6 marks)

6. *(from 1.1.1)* State the role of the **Program Counter (PC)** and the **Memory Address Register (MAR)** during the fetch stage of the fetch–decode–execute cycle. **[2]** *(AO1)*

7. *(from 1.1.1)* Explain how **increasing the number of cores** in a processor can improve performance, and give **one** reason why doubling the cores does not usually double real-world performance. **[2]** *(AO2)*

8. *(from 1.1.3)* State **one** advantage of a **solid state drive (SSD)** over a magnetic **hard disk drive (HDD)**, and **one** advantage of an HDD over an SSD. **[2]** *(AO1)*